TRADE REPUBLIC

RECRUITING

DATA SCIENCE CHALLENGE

TRADE REPUBLIC

To proceed with your interview process at Trade Republic, we have prepared a short data science challenge for you. The goal of the challenge is for you to show that you are an expert in your field and committed to bring the data team at Trade Republic to the next level.

Please describe briefly what you have done and where you think your solution could be developed further, why you have decided for some ways and what you would like to work on/ analyze, even though it was not a direct part of the requirements.

Part 1: Advanced SQL

In the first part, we ask you to provide some SQL queries for certain questions from the business side which should be answered.
Every trade is stored in a database, where the user, the execution timestamp, the traded instrument and the number of shares, execution price and execution total are persisted.

Table: **trade**

user_id	timestamp	instrument_id	quantity	price	total
2342	2019-03-21 15:03:27	DE000BASF111	25.00	53.70	1342.50
…					

Please provide a single SQL query (can be nested), which returns as columns the date, the daily, weekly (rolling 7-day window) and monthly (rolling 30-day window) trading users for the last 30 days from the current day. Specifically, we want to create a table of the following form:

Table: **active_users**

date	daily_trading_users	weekly_trading_users	monthly_trading_users
2019-03-21	116	250	301
2019-03-20	110	243	292
…			

Extend the query to also indicate trading volume for each day (sum of all trade totals of that day).

Which ways exist in general in SQL to – within one query – execute a sub-analysis for each result row? E.g. one part of the query returns a result set of 30 rows and for each row we want to execute another analysis.

Recommended time: 30 min

Part 2: Analyze trading activity data

For the second part we provide you with some data, covering a set of six fictitious customer cohorts that started trading in H1 2020. It consists of three csv files: *trades*, containing data on executed trades, *prices*, the respective end of day prices, and *instruments*, which contains data on the index the instrument belongs to, as well as its type, country, and sector..
To analyze the data, please use your scripting language and frameworks of choice, e.g. R or Python.

From the trading data, some insights can be calculated. We want to generate a daily report with these insights showing a few visualizations for these cohorts.

We are most interested in the following metrics over time:

- Cumulative customers
- Number of trades
- Volume of trades
- Average trade size
- Assets held by customers (at current value)

What other metrics would you look at to understand if these customers perform well? Keep in mind that we are making money from trades, but are more interested in a long-term stable customer relationship than excessive short-term gains.

Please don't spend too much time here, that's mostly to familiarize yourself with the data and get you warmed up for part 3.

Recommended time: 60 min

Part 3: Cluster customers

Let's dig a bit deeper into the data and understand what different customers we have by segmenting them into different groups:

Calculate some basic features for each customer:

- Monthly volume, trades, and respective avg. trade size
- Performance - gains of each customer over time:
  - realized profits, where there is a sell trade following the buy, and
  - virtual profits, where there is no corresponding sell trade yet (value the open position at the last end of day price)
- Distinct number of instruments they traded, and distinct number of sectors

What other features could be useful? Some initial ideas:

- Can we further classify instruments based on our data?
- Are there patterns in trading times?
- Portfolio concentrations?

Cluster the customers into several groups. Please comment on your decisions and assumptions, e.g. how you decided on the number of groups.

Please prepare all your results in a form presentable to the management board. Ideally add some action items you would suggest.

Recommended time: 150 min

Bonus:

- *Situation:* We want to know early on in the customer lifecycle what we can expect from a customer later on, to make sure our marketing campaigns have a reasonable ROI soon after we launched them.
- *Task:* Based on the characteristics of the first 3 trades of a customer, can we predict how many trades a customer will do in their 6th month after

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Finalization

Please submit your solution with all required files via email.

We are looking forward to the further discussions!